

Statement of Capabilities

U.S. Geological Survey (USGS)

Earth Resources Observation and Science (EROS) Center

1. Introduction

The U.S. Geological Survey's Earth Resources Observation and Science (EROS) Center provides secure, resilient, and scalable hosting services designed to meet the operational and continuity needs of federal programs, while aligning with USGS and Department of the Interior (DOI) policy frameworks.

The EROS Center delivers disciplined, standards-based Information Management Technology services that enable federal missions, scientific workloads, and enterprise applications.

2. EROS Facility Capabilities (Facilities as a Service – FaaS)

EROS offers a fully managed, enterprise-grade physical facility environment for collocating government IT systems. The current backup LMOC (bLMOC) is hosted at EROS.

2.1 Facility Infrastructure

- Secure, conditioned rack and floor space
- Clean, stable electrical power with UPS bridging and generator-backed failover
- Precision HVAC ensuring consistent cooling, humidity control, and air quality
- Environmental monitoring for critical infrastructure

2.2 Physical Security

- 24×7 onsite physical security staffing
 - Secure computer room access
 - Continuous video monitoring
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3. Virtual Infrastructure Capabilities (Infrastructure as a Service – IaaS)

EROS provides scalable virtual compute resources enabling customers to host applications within a secure, managed federal environment.

3.1 Core Capabilities

- Virtual machines sized per customer-defined CPU, memory, and storage requirements
- Modern, refresh-cycled virtualization hardware for performance and stability
- High-performance enterprise storage with solid-state and hybrid options
- Redundant compute and network paths to ensure service availability
- Standardized VM templates enabling secure, consistent deployments

3.2 MOC Vendor Responsibilities

MOC Vendors maintain control over:

- Applications and workloads deployed on the VM
- System-level configuration and tuning to support mission requirements

3.3 EROS (GFE)-Provided Support

- Standardized snapshots and backup support as required
 - Resource adjustments (CPU, memory, storage) upon request
 - Base VM provisioning within established timeframes
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4. Network & Connectivity Capabilities

EROS maintains a robust enterprise network environment engineered to support mission, enterprise, and scientific workloads.

4.1 Core Network Capabilities

- Redundant internal and external network paths for high availability including proven network capabilities for secure, remote MOC operations
 - Secure segmentation and traffic isolation to protect federal systems
 - Support for high bandwidth operations and scientific data movement
 - Integration with federal backbone and research-focused network services
 - Network monitoring designed to protect against threats and ensure operational continuity
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5. Software Defined Networking (SDN) Capabilities

EROS leverages modern, software-defined network technologies as part of its enterprise network strategy. These capabilities provide adaptability, enhanced security, and performance improvements for workloads.

EROS employs software-defined networking solutions that enable:

- Support for Federal Zero Trust Network Access (ZTNA) requirements
- Centralized management of network policies and routing
- Rapid provisioning and modification of network segments to support project needs
- Improved visibility and control across distributed network environments
- Dynamic traffic optimization to improve performance and reduce bottlenecks

These SDN capabilities enhance overall network performance, scalability, and security—supporting mission-critical systems and evolving federal IT requirements.